Dr Jennifer L. Hawkin
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Academic Experience

2024-present **Postdoctoral Researcher**, *University of Sheffield, UK (School of Chemical, Materials & Biological Engineering)*, Resource Efficiency: Plastics and Chemical Systems, **Supervisors**: Professor A.J. Ryan (FRSC), Dr F. Meng (MRSC)
Working closely with colleagues in the School of Mathematical and Physical Sciences (Chemistry) to evaluate waste management and circularity associated with petrochemical products and energy infrastructure (see publications section for more details).

2021-2024 **PhD**, *University of Cambridge, UK (Engineering Dept.)*, **Supervisor:** Professor Julian Allwood, **Title:** Do Net-Zero plans add up? A framework and model to quantify risks of resource supply shortages in climate mitigation strategies

2012-2021 **Industrial Experience (see below)**, *Rolls-Royce & Engineers-Without-Borders* UK, Singapore, Malaysian Borneo

2008-2012 **BA & MEng (Hons) Mechanical Engineering (focus area: fluid dynamics)**, *University of Cambridge, UK*, Distinction
Included one year studying at Massachusetts Institute of Technology (M.I.T.), USA

Industrial Experience

2017-2021 **Rolls-Royce PLC, UK**, *Roles included: Technical Assistant to the Chief Technology Officer, Sub-System Integration & Component Design, Innovation Technology Lead.*
- Diverse high-profile responsibilities against demanding timescales
- Developed an insight into the reality of initiating change in large industrial companies

2016-2017 **TONIBUNG (Friends of Village Development), Malaysian Borneo**, *EWB-UK (Engineers-Without-Borders UK) Fellowship Volunteer*
- Collaborated with local and international colleagues to develop and roll out novel technology for clean electricity generation in remote communities

2016 **Rolls-Royce, Singapore**, *Regional Coordinator (Secondment)*
- Scoped and drafted the proposal for a government funded research centre in Singapore

2012-2016 **Rolls-Royce PLC, Derby/Bristol/Singapore**, *Various roles in Supply-Chain and the Strategic Research Centre*
- Projects included computational and experimental fluid flow characterization and cross-sector work, taking a strategic view of emerging technologies.

Teaching Experience

2025-2026 **Project Co-Supervisor**, *University of Sheffield*, Masters-level student projects
- Comparative Life Cycle Analysis of Elium and Conventional Epoxy Systems for Turbine Blade Manufacturing
- Assessing the future increased polymer demand in the transition to domestic Battery Electric Vehicles within the UK

2025 **Tutor**, *University of Cambridge (Engineering)*, Masters level: "Climate Change Mitigation"

- Providing guidance on student proposals for climate mitigation options. Student projects must consider technical potential of proposals and rates of deployment within the whole system of global emissions. Student projects span all economic and industrial sectors.

2022-2024 **Facilitator**, *University of Cambridge (Engineering)*, Undergraduate poster sessions: "Sustainable Engineering"

- Facilitating discussions on sustainable engineering. Student topics may consider any aspect of sustainability and often span all economic and industrial sectors.

2021-2024 **Tutor**, *University of Cambridge (Engineering)*, Masters and undergraduate level: "Climate Change Mitigation" and "Structural Design"

2022-2023 **Supervisor**, *St Catherine's College, University of Cambridge*, Second-year undergraduate course: Materials thermodynamics & diffusion and materials processing

2022-2023 **Interviewer**, *St Catherine's College, University of Cambridge*, Undergraduate admissions (Engineering)

Selected Conference and Event Participation

2025 **Oral & Poster Presentations**

- "Re-evaluating plastics end-of-use treatment options" and "Engineering Risk Assessments for Climate Policy - Can Failure Modes and Effects Analysis (FMEA) improve the prospects of Carbon Dioxide Removal (CDR)?", presented at *the Industrial Society for Industrial Ecology (ISIE) Conference (Singapore)*
- "Do Net-Zero plans add up?", presented at *the World Resources Forum (online)*
- "Constraints on carbon circularity in the petrochemical sector", presented at *the Annual EDITS Meeting (Energy Demand changes Induced by Technological and Social innovations), IIASA, Laxenburg, Austria*

2024 **Oral Presentation**, *Annual EDITS Meeting (Energy Demand changes Induced by Technological and Social innovations)*, IIASA, Laxenburg, Austria
"A model and interactive calculator to aggregate demands of climate mitigation proposals"

Invited participant, *UNEP & Life Cycle Initiative Biogenic Carbon in LCA Guidance Project Workshop*, Bath, UK

Poster, *ISIE (Industrial Society for Industrial Ecology) Gordon Research Conference*, Les Diablerets, Switzerland
"Is Net-Zero at Risk? Modelling aggregated energy demands of climate mitigation plans reveals high risk 2050 expectations."

Invited discussion leader, *ISIE Gordon Research Seminar*, Les Diablerets
"Decarbonization Across Sectors"

2023 **Invited speaker**, *Panel discussion: Clearing the air - information literacy in the climate crisis*, Cambridge University

Oral Presentation, *Symposium: "Governance, law and economics of climate change and energy transition"*, Cambridge University
"What scale of negative emissions can we rely on?"

Oral Presentation, *ISIE Conference*, Leiden, The Netherlands
"A framework for aggregating energy demands of net-zero proposals"

2022 **Oral Presentation**, *Cambridge Zero Research Symposium*
"Should 'we' change our behaviour to reach net-zero?"

2017 **Invited speaker**, *Institution of Mechanical Engineers, Singapore Network*
"Micro-Hydro Power - Clean and Affordable Electricity in the Jungle"

2015 **Invited speaker**, *Institution of Engineering and Technology East Midlands Network*
"Are Unmanned Autonomous Vessels the Future for Commercial Shipping?"

Professional Development and Institutional Engagement

2025 **Prize: Best Poster (runner-up)**, *ISIE Conference (Singapore)*

2022-2024 **Equity, Diversity & Inclusion Steering Group**, *Darwin College, Cambridge*

Additional Skills

Coding:
- Python for model implementation
- RStudio/RMarkdown for documentation

Languages:
- Conversational Italian
- Conversational German
- Basic French

Other activities and interests:
Recreational sports (including swimming, cycling, rambling and rock climbing); European folk dance; Scouting

Publications

2026 **Aggregating demand for three fundamental resources to avoid burden-shifting in climate policy**, *Environmental Science & Technology*, **Hawkin, J.L.**, & Allwood, J.M., DOI: 10.1021/acs.est.5c12742
By aggregating demand for emissions-free electricity, biomass and carbon storage required by climate policy proposals across sectors, we show they imply implausible deployment rates, unless demand is reduced through participatory innovations.

Reducing greenhouse gas emissions from polymer processing at end-of-use, *Journal of Cleaner Production*, Gao, Y., Meng, F., **Hawkin, J.L.**, Cullen, J., & Cabrera Serrenho, A., DOI: 10.1016/j.jclepro.2026.148308
We quantify greenhouse gas emissions associated with end-of-use options for polymers, globally. Our findings reveal that enhancing recycling and landfilling could substantially reduce emissions by 30%, while mismanagement practices such as open burning and dumping exacerbate environmental pollution and must be avoided.

Re-evaluating plastics end-of-use treatment options for carbon sequestration (under review), *Nature Reviews Materials*, **Hawkin, J.L.**, Ryan, A., Cullen, J., & Meng, F.
We compare plastic waste treatment options, including mechanical recycling, advanced recycling, waste-to-energy technologies, conventional landfill, and a hypothetical "curated storage". We find that treatment pathways require more nuanced evaluation, including consideration of feedstock composition, chemical reactions/retrosynthesis, process and infrastructure maturity, energy system context, governance constraints, and long-term liability.

Rethinking circularity in chemicals: why targeting carbon loops is not enough (under review), *Chem Circularity*, **Hawkin, J.L.**, & Meng, F.
This paper responds to rapidly expanding net-zero and circular economy strategies that increasingly prioritise "carbon circularity" within the chemicals sector, such as recycling, alternative feedstocks, and carbon capture and utilisation. Rather than assuming that closing carbon loops will deliver transformative mitigation at scale, we synthesise evidence across deployment feasibility, energy and resource constraints, and circularity hierarchies to evaluate what these strategies can and cannot deliver in practice.

Science-based targets to balance environmental impacts and inequalities of material production and use (in preparation), **Hawkin, J.L.**, Watari, T., Ryan, A., & Meng, F.
Drawing on existing scientific frontiers, preliminary Science-based targets are proposed for four key materials: steel, cement, plastics, and sand.

2025 **End of life management for wind turbines**, *Nature Reviews Clean Technology*, Meng, F., **Hawkin, J.L.**, Gast, L. *et al.*, DOI: 10.1038/s44359-025-00097-3
This Review assesses current and emerging EOL practices, comparing the environmental and economic trade-offs across mechanical, thermal and chemical recycling technologies to recover materials from wind turbines.

Treatment of temporary GHG removals in voluntary carbon markets, *Cambridge Open Engage*, Zschietzschmann, H., **Hawkin, J.L.**, Rosser, J.P. *et al.*, DOI: 10.33774/coe-2025-5sbl5
This report provides a broad introduction to the issue of carbon storage permanence, equivalence and accounting approaches.

2024 **Do net-zero plans add up? A framework and model to quantify risks of resource supply shortages in climate mitigation strategies**, *PhD Thesis*, **Hawkin, J.L.**, DOI: 10.17863/CAM.119203
This thesis develops a framework to aggregate resource demands of net-zero proposals by modelling low-carbon and conventional processes across industry and domestic sectors. Application of the model shows that dominant mitigation strategies rely on implausible deployment rates of non-emitting electricity generation and carbon storage and/or unsustainable use of biomass.